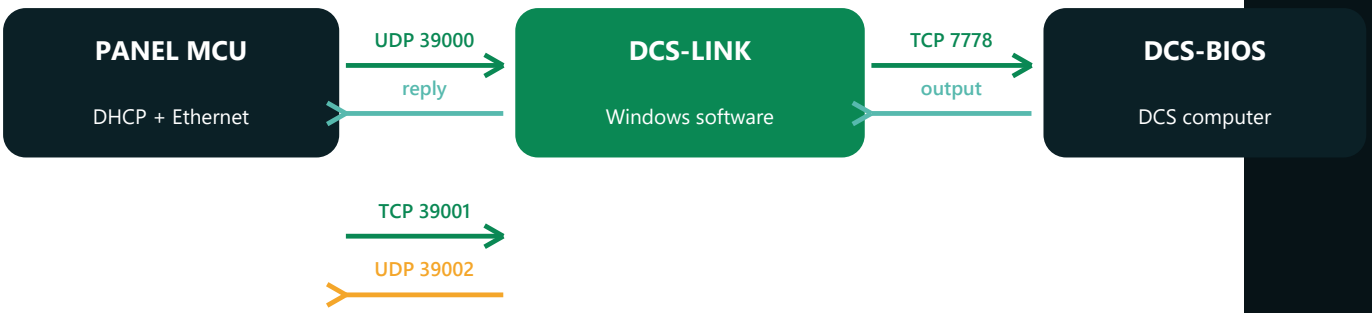
Software and Network Troubleshooting Guide

Avionicscraft Simulator Software

Purpose
A practical guide for diagnosing DCS-LINK, DCS-BIOS, Windows Firewall, DHCP and Ethernet-connected panel MCU problems.



Panel discovery and DCS-BIOS use separate connections

DCS-LINK | DCS-BIOS | ETHERNET MCU PANELS

Revision: 13 November 2026

SOFTWARE / NETWORK TROUBLESHOOTING

1. Start here

Two independent connections
The DCS-BIOS connection and panel connections are separate. A green DCS-BIOS indicator does not prove that a panel is connected. A panel can also appear under Connected Modules while the indicator is orange.

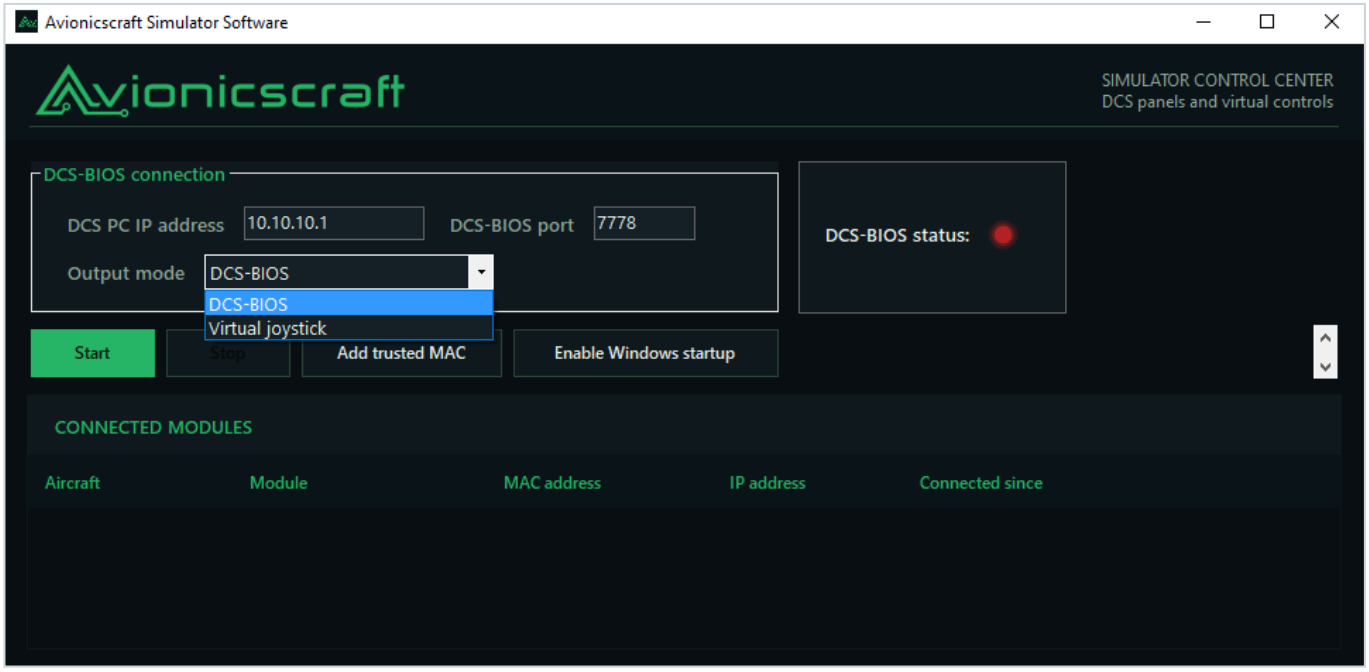


Figure 1. Example of the Avionicscraft Simulator Software interface. The activity-log area is intentionally excluded.

CONNECTED

DCS-BIOS TCP link is active

RETRYING

DCS-LINK retries every 3 seconds

STOPPED

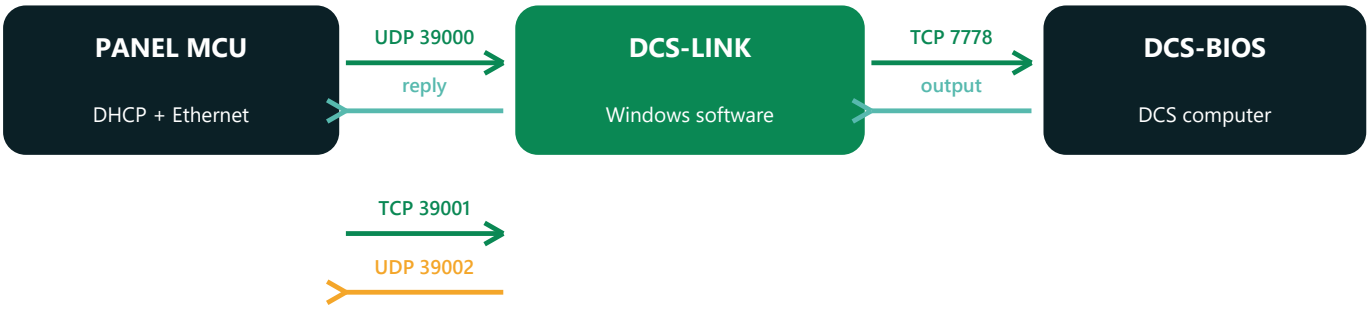
Click Start to enable DCS-LINK

Quick diagnosis

Symptom	First check
Status is red	Click Start. Red means DCS-LINK is stopped.
Status remains orange	Check the DCS computer address, TCP 7778 and DCS-BIOS.
Green status, no panel	Check DHCP, UDP 39000, TCP 39001 and the LAN.
Panel appears, then disappears	Check periodic MCU traffic, link stability and duplicate MAC addresses.
TCP entry, no panel row	Check the identity line and trusted-MAC allow-list.
Inputs visible, DCS unchanged	Check DCS-BIOS control identifiers and numeric values.
LED test fails	Check UDP 39002, PWM polarity, common ground and the driver.

2. How the connections work

The software uses four local-network paths. Understanding which path has failed prevents unnecessary changes elsewhere.



Panel discovery and DCS-BIOS use separate connections

Port	Protocol	Direction	Purpose
7778	TCP	DCS-LINK to DCS-BIOS	Commands and simulator output stream
39000	UDP	MCU to DCS-LINK	Automatic panel discovery
39001	TCP	MCU to DCS-LINK	Persistent panel connection
39002	UDP	DCS-LINK to MCU	Commands such as LED testing

Local network only
Do not forward these ports to the internet. They are intended for a trusted private LAN. Discovery broadcasts normally do not cross routers or isolated VLANs.

Recommended diagnostic order



- Confirm power and an active Ethernet link.
- Confirm that the MCU receives a valid DHCP address.
- Start DCS-LINK and confirm that its listeners open.
- Read the activity log for identity or trust errors.
- Confirm that the MCU and computer share the same LAN or VLAN.
- Test TCP 7778 independently from panel discovery.

3. DCS-LINK will not start

UDP 39000 or TCP 39001 is already in use

Only one active DCS-LINK instance can listen on the discovery and panel ports. Close other copies and click Start again.

```
Get-NetUDPEndpoint -LocalPort 39000 -ErrorAction SilentlyContinue
Get-NetTCPConnection -LocalPort 39001 -State Listen -ErrorAction SilentlyContinue
```

Use the returned OwningProcess value to identify the program:

```
Get-Process -Id <PROCESS_ID>
```

> Safety note
> Do not terminate an unknown process until it has been identified. The normal fix is to close an extra copy of the simulator software.

The application does not open

- Use Windows 10 or Windows 11.
- Install the .NET 6 Desktop Runtime for a framework-dependent build.
- Check whether antivirus or Windows SmartScreen quarantined the executable.
- Open Event Viewer > Windows Logs > Application and inspect the launch-time error.
- Try a fresh Release build or a self-contained Windows x64 publication.

Listener health

After recoverable network errors, the UDP discovery and TCP panel services restart automatically after approximately one second. A normal temporary adapter interruption should not require restarting the application.

4. A panel is not discovered

Physical network and DHCP

- Verify power, PoE where applicable, the cable, link LEDs and switch port.
- Confirm that the DHCP server issued a valid IPv4 address.
- An address beginning with 169.254 usually indicates that DHCP was unavailable.
- Give every MCU a unique MAC address.
- Check for duplicate static IP addresses if any device still uses manual addressing.

Same broadcast domain

Automatic discovery uses local UDP broadcast traffic. The MCU and DCS-LINK computer should normally share the same IPv4 subnet and broadcast domain.

Possible blocker	What to check
Guest or client isolation	Move both devices to a trusted, non-isolated network.
VLAN rules	Allow local broadcast and the required TCP/UDP paths.
VPN	Temporarily disconnect it while testing routing.
Multiple adapters	Inspect Ethernet, Wi-Fi, Hyper-V and virtual-machine routes.
Public firewall profile	Use Private only on a trusted cockpit LAN.

Inspect the computer configuration:

```
ipconfig /all
Get-NetConnectionProfile
Get-NetIPAddress -AddressFamily IPv4 | Select InterfaceAlias,IPAddress,PrefixLength
```

MCU retry requirements

- Broadcast discovery repeatedly until DCS-LINK responds.
- Reconnect TCP 39001 after a lost connection.
- Send a valid identity line immediately after connecting.
- Continue sending state or heartbeat traffic while connected.

5. Activity-log decoder

The log usually reveals the last successful stage. Work from the earliest unexpected message rather than changing every network setting at once.

Log entry	Meaning and action
No panel-related entry	Discovery probably did not reach DCS-LINK. Check subnet, firewall and repeated MCU discovery.
Panel TCP connection from ...	UDP discovery succeeded and TCP 39001 opened. DCS-LINK now expects identity.
did not send its identity within 5 seconds	The MCU connected but did not send identity quickly enough.
sent an invalid identity	Check protocol format, MAC format and newline termination.
Unknown panel rejected	The temporary trusted-MAC allow-list is active.
Panel rejected because its MAC address is not trusted	Add this MAC or restart DCS-LINK to clear the temporary list.
Panel connected	Discovery, TCP and identity validation succeeded.
Panel disconnected	The TCP socket closed. Check MCU reconnect logic and link stability.
Module connection lost	No MCU data arrived for more than five seconds.
DCS-BIOS is unavailable	Check the DCS PC address, TCP 7778, service and firewall.

Trusted MAC behaviour
The allow-list exists only for the current DCS-LINK session. With no entries, all valid MAC addresses are accepted. After one MAC is added, every other MCU must also be added. Stop and Start clears the temporary list.

Connected and then lost

DCS-LINK removes a panel after five seconds without data. A complete state snapshot every 100 ms provides responsive monitoring and stays well within this timeout.

6. DCS-BIOS remains unavailable

Orange means that DCS-LINK is running and retrying. It reconnects approximately every three seconds.

DCS-BIOS on the same computer

- Enter 127.0.0.1 as the DCS PC IP address.
- Keep port 7778 unless the configuration explicitly uses another port.
- DCS-BIOS can start before or after DCS-LINK; retrying is automatic.

```
Test-NetConnection -ComputerName 127.0.0.1 -Port 7778
Get-NetTCPConnection -LocalPort 7778 -State Listen -ErrorAction SilentlyContinue
```

DCS-BIOS on another computer

- Enter the DCS computer's LAN IPv4 address, not 127.0.0.1.
- Allow inbound TCP 7778 on that computer's Private firewall profile.
- Confirm that the service listens on the LAN address or all interfaces.

```
Test-NetConnection -ComputerName <DCS_PC_IP> -Port 7778
```

Reading the result
TcpTestSucceeded: False indicates a listening-service, routing, firewall or address problem. True proves that a TCP service accepted the connection, but not that it supplies valid DCS-BIOS data.

Important distinction

An orange DCS-BIOS indicator does not stop MCU discovery. If panels are also missing, troubleshoot the panel LAN path separately.

7. Windows Firewall

When Windows asks, allow Avionicscraft Simulator Software on Private networks. If the prompt was dismissed, open:

> Windows path
> Windows Security > Firewall & network protection > Allow an app through firewall

- Allow inbound UDP 39000 and inbound TCP 39001 on the DCS-LINK computer.
- Outbound TCP 7778 and UDP 39002 are normally allowed by default.
- If DCS-BIOS runs elsewhere, allow inbound TCP 7778 on that computer.
- Update the firewall rule if the executable was moved to another folder.

Inspect application rules

```
Get-NetFirewallRule | Where-Object DisplayName -Like '*Avionicscraft*' |
    Select DisplayName,Enabled,Direction,Action,Profile
```

Optional inbound rules

Run these commands as Administrator only when no suitable application rule exists:

```
New-NetFirewallRule -DisplayName 'Avionicscraft UDP discovery' `
  -Direction Inbound -Protocol UDP -LocalPort 39000 -Action Allow -Profile Private

New-NetFirewallRule -DisplayName 'Avionicscraft panel TCP' `
  -Direction Inbound -Protocol TCP -LocalPort 39001 -Action Allow -Profile Private
```

> Use Private profile only
> Do not open cockpit-control ports on an untrusted public network and do not create internet port-forwarding rules.

8. A panel connects and disconnects

Observed behaviour	Likely cause
Identity timeout	Identity was not sent immediately after TCP connected.
Connection lost after about 5 seconds	No periodic state or heartbeat data was received.
Immediate Panel disconnected entry	The MCU or network closed the TCP socket.
Same MAC reconnects repeatedly	Duplicate MAC, duplicate identity or unstable firmware.
All panels drop together	Switch, router, PC adapter, power or PoE interruption.

Keep one persistent connection
The MCU must keep TCP 39001 open. Do not reconnect for every switch event. After any socket failure, repeat discovery and establish a new persistent connection.

Unique MAC addresses

DCS-LINK identifies panel connections by MAC address. A newly connected duplicate replaces and closes the older connection. DHCP servers can also behave unpredictably when two MCUs use the same MAC.

Useful serial messages

Open the MCU serial monitor at the firmware's configured baud rate and look for a progression similar to:

```
Requesting DHCP address...
DHCP address: <MCU_IP>
DCS-LINK found: <PC_IP>:39001
Connected to DCS-LINK.
```

Repeated DHCP failures, discovery attempts or TCP failures identify the stage requiring attention.

9. Controls, outputs and mission sync

Controls do not affect DCS

Check	Expected result
DCS-BIOS status	Green
Connected Modules	The panel is listed and remains present
View module status	Input values change when controls move
Command format	CONTROL_IDENTIFIER VALUE followed by a newline
Identifier	Exactly matches the selected aircraft's DCS-BIOS control
Value	Numeric and valid for that control
Wiring	Switch common and MCU ground are correct

Mission input synchronization

Input telemetry must identify the component as INPUT and provide a numeric value from 0 through 65535. DCS-LINK sends stored input states when DCS-BIOS connects and repeats them approximately once per second.

- Wait for the DCS-BIOS indicator to become green.
- Confirm that the current physical state is visible in the module status window.
- Match the component identifier exactly to the DCS-BIOS input identifier.
- Continue transmitting complete input snapshots after mission loading.

DCS output or lighting does not work

- Keep reading the TCP stream even when no physical input changes.
- Verify the DCS-BIOS output address and parser configuration.
- Check PWM GPIO, frequency, resolution and active polarity.
- Use a common ground and a suitable external LED driver.
- Never power an LED strip directly from an MCU GPIO.

LED test result
The test command uses UDP 39002. If testing works but DCS-controlled lighting does not, the network path and output hardware probably work; inspect the DCS-BIOS output mapping. Opposite brightness behaviour usually indicates active-low drive polarity.

10. Module status, startup and settings

Module status does not update

- Send PANEL_STATE messages over the existing TCP connection.
- End every message with a newline.
- Keep component identifiers stable.
- Use numeric values for inputs that participate in mission sync.
- Confirm that the Updated time changes in the status window.
- If the panel disappears after five seconds, periodic traffic is not arriving.

Windows startup

Enable Windows startup creates an entry for the current Windows user and automatically starts DCS-LINK after sign-in.

- Disable and re-enable startup after moving or renaming the executable.
- Confirm that the executable still exists at the registered path.
- Check Task Manager > Startup apps and Windows security notifications.
- Start manually and verify the saved DCS address and port.

Reset damaged settings

Settings are stored in:

```
%LOCALAPPDATA%\AvionicscraftSimulatorSoftware\settings.json
```

Close the application, then preserve the old file as a backup:

```
Rename-Item "$env:LOCALAPPDATA\AvionicscraftSimulatorSoftware\settings.json" `
    "settings.json.bak"
```

Activity log
The log exists only in memory. Copy useful entries before closing the application. Increase Keep entries before reproducing an intermittent problem. Clear log does not affect network connections.

11. Advanced diagnostics and support

Wireshark packet capture

Capture the active Ethernet adapter and apply this display filter:

```
udp.port == 39000 || tcp.port == 39001 ||
udp.port == 39002 || tcp.port == 7778
```

Expected panel sequence

Step	Expected network event
1	MCU broadcasts discovery to UDP 39000.
2	DCS-LINK replies with the TCP panel port.
3	MCU opens TCP 39001.
4	MCU sends its identity line.
5	MCU sends ongoing state or heartbeat traffic.
6	DCS-LINK connects independently to DCS-BIOS on TCP 7778.

Capture traffic only on computers and networks you are authorized to inspect.

Information to collect for support

Category	Information
Software	Build or release number, Windows version
DCS-BIOS	Same PC or remote PC, configured IP address and port
Status	Indicator colour and relevant activity-log entries
Network	Computer IPv4/prefix, MCU IPv4 and unique MAC
Scope	Whether other panels connect successfully
Firmware	Serial output from startup through the failure
Test	Result of Test-NetConnection for TCP 7778

Support website
https://avionicscraft.com/